

Bar inventory forecasting

Take-home report | six bars, 16 brands | historical movement log through 1 Jan 2024

1. What is the core business problem and why does it matter?

A bar can lose sales and disappoint guests when popular brands run out, while excess stock ties up cash and raises waste or shrinkage risk. The decision is the target stock for each **bar-brand**, not one target across the chain. The prototype forecasts logged usage in ml, proposes an illustrative par level, and shows how a manager could review replenishment. It does not place purchase orders.

2. What assumptions did you make? Why?

The 6,575 records have irregular bar-brand observation intervals (median 3.86 days), so a row's consumed ml is treated as usage **since the previous observation**. First intervals are excluded because their duration is unknown; missing dates are not called zero demand. This assumes the logs record accumulated movement, suggested by opening and prior closing balances usually matching, but the hotel must confirm it. There are 86 rows with balance identity gaps over 1 ml and 129 transitions with opening gaps over 1 ml. Consumption may understate demand during stockouts. No current counts, open orders, supplier terms, case sizes or costs are supplied.

3. Which model, and why not others?

A duration-weighted lifetime daily usage rate won the validation comparison against a recent-six-interval rate and a 30-day-prior shrunk blend. Validation MAE was **245.79 ml per observed interval** for lifetime, versus 248.38 and 276.30 ml. The sample is sparse (40-99 rows per bar-brand) and irregularly timed, so transparent rates are more defensible here than seasonal ARIMA, gradient boosting, or deep learning without reliable daily sales, occupancy and stockout labels. A training-only 99th-percentile cap limits anomalously high interval rates. Forecasts update after each known observation; they do not see future consumption.

4. How does it perform, and what would improve it?

Chronological split: train before 1 Oct 2023, select on 1 Oct-14 Nov, test from 15 Nov 2023 through 1 Jan 2024. On **852 untouched test intervals**, the selected model has **MAE 247.80 ml/interval**, WAPE 82.97%, and mean actual-minus-predicted bias -1.74 ml. This high error means no reliable service level or stockout savings can be claimed. Next: collect daily POS sales, on-hand counts, stockout flags, events, occupancy, losses and actual lead times; reconcile bad movements, evaluate fixed-horizon forecasts and compare against current ordering policy using backtests.

How it becomes an operating decision

Recommendations are scenario outputs, not live orders.

5. How would this work in a real hotel?

Each morning, ingest yesterday's POS movement, receipts, transfers, waste, physical count and outstanding purchase orders per bar-brand. Refresh the daily forecast; choose a stock target for lead time plus review cycle; subtract **current on-hand + open orders** to suggest the order, then apply supplier pack sizes and manager approval. As a demonstration, the notebook uses an assumed 2-day lead time, 1-day review, 20% provisional buffer and rounds up to 100 ml: $\text{par} = \text{ceil}(\text{forecast ml/day} \times 3 \times 1.20 / 100) \times 100$. These parameters are examples, not hotel policy or a calibrated service guarantee. The last closing balance in the supplied sheet is from 2023-24, so its suggested order amounts must never be treated as current.

What would break at scale, and what should be tracked?

Sparse or delayed logs, stockouts that censor sales, unrecorded transfers, inconsistent brand IDs, late suppliers and different bottle sizes can invalidate the suggestion. Version the model and input data, alert on missing feeds, show data freshness and allow human overrides. Track forecast MAE/WAPE and bias per bar-brand, stockout days, unmet demand if measurable, days of inventory, waste/shrinkage, slow stock, lead-time drift and override rates. Recalibrate reorder buffers against actual service targets and carrying costs.

Simulation explained

The notebook runs a seven-day synthetic scenario for one bar-brand. It starts from a *hypothetical* 150 ml stock count, uses forecast-level demand with a weekend-like spike, receives orders after two days, and records arriving, ordering, served, unserved and closing ml. It yields 8 ml of unserved demand in this toy example; this is not an observed hotel result or a measured improvement.

Source and reproducibility

Source: [supplied Google Sheet](#), CSV snapshot downloaded 28 Sep 2026. The executed notebook and README contain the audit, chronological evaluation, all 96 illustrative recommendations and simulation. The form separately requires a self-recorded 3-5 minute video.